NMB	Rod End Bearing Division	Procedure	
Establish Date : 11 / SEP / 24	Engineering Design for Sleeve	Document No.	PD-DV-EN-121003
Revision: [ ]		Nanthiwa. / Prepared by	Checked by / Approved by
Effective Date : 16 / SEP / 24		Karuizawa Approved by Y. Hidano	
		Occupation (ENG)	

1. SCOPE

   This procedure is prescribed for the design process and dimension of sleeve at the ROD END Bearing division.

2. PURPOSE

   This procedure for design dimension of the sleeve part and process for producing the part according to customer requirement.

3. DEPARTMENT IN CHARGE

   Engineering Department.

4. REFERENCE SPECIFICATION

PD-DV-EN-111002	Procedure for Feasibility and Specification Review
PD-DV-EN-111003	Procedure for Process Flow Diagram (PFD)
PD-DV-EN-121007	Engineering Design of Liner cutting
PD-DV-EN-121011	Engineering Standard : Unit Conversion
PD-DV-EN-121013	Engineering design for plating tolerance
PD-DV-QC-230000	Control of Non-conformance
RE23018	Manufacturing design manual of sleeve

5. DEFINITION

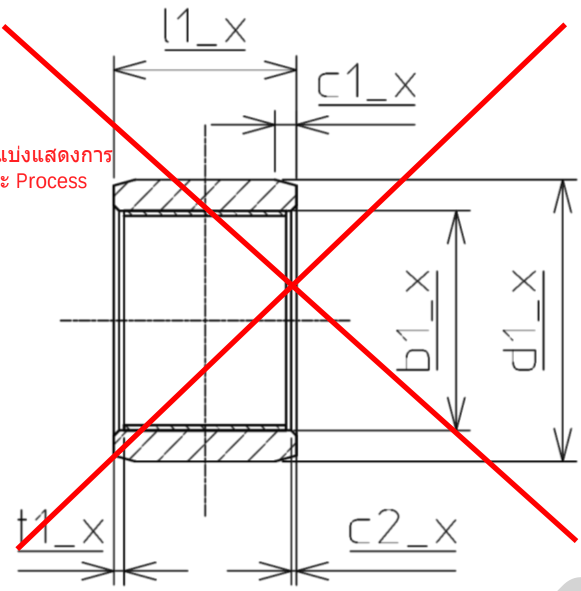
   5.1 Upper : Upper tolerance of spec
   5.2 Lower : Lower tolerance of spec
   5.3 Unit : Units of dimension are shown in millimeter (mm.) and degree (°).
   5.4 Outer diameter : Outside diameter of sleeve
   5.5 Inner diameter : Inside diameter of sleeve without liner
   5.6 Chamfer of Outer diameter : Chamfer at edge on the outside diameter of sleeve
   5.7 Chamfer of Inner diameter : Chamfer at edge on the inside diameter of sleeve
   5.8 Length of Sleeve : Distance between flat face to flat face of sleeve
   5.9 Setback : Distance between end of liner with end of sleeve length
   5.10 GD&T : Geometric Dimensioning and Tolerancing

6. PROCESS DESIGN FOR SLEEVE

   6.1 In case new model specification or calculated specification are the same similar model it can be referenced to a similar model. If not matched, the design will follow this procedure.
   6.2 Process design for Sleeve shall be designed from customer requirement to internal process because some processes have an effect on the dimension of the product. The flow of process design shall be referenced on $\boxed{\text{PD-DV-EN-110003}}$ Procedure for PFD
   PD-DV-EN-111003

7. DIMENSION DESIGN OF SLEEVE

ไม่ใช้รูปแบบนี้แต่จะแบ่งแสดงการ
คำนวณเป็นของแต่ละ Process



แก้คำนิยามของ Final dimension ไม่ต้องคำนวณเป็นค่ากลาง แต่ให้แสดงเป็น tol. ตามที่ Drawing spec ระบุ
เว้นแต่ว่าค่าใน drawing spec ระบุไว้เป็น MIN-MAX ให้คำนวณเป็นค่ากลาง

7.1 Final dimension

Final dimension with tolerance from Customer Specification use on Final Inspect should be transformed to mean dimension between upper and lower tolerance. Some dimension absence specified from customer specification shall be referred with a similar model first, If not specified shall follow this procedure. The dimension not matched with customer specification shall refer to the customer specification.GD&T designations will use _gt_1, _gt_2,..., _gt_x respectively at the end of the relevant dimension name refer customer specification or internal document required.Other dimension if not specified should be refer from customer specification. ตรงส่วนนี้ไม่แน่ใจว่าต้องการให้ระบุกับอะไร

7.1.1 Outer diameter (ØD)

$$d1\_1 = \frac{(ØD + upper) + (ØD + lower)}{2}$$

7.1.2 Inner diameter

$$b1\_1 = [\frac{(ØB + upper) + (ØB + lower)}{2}] + t$$

$$\boxed{t = liner\ thickness\ \ (X - 1820\ \ =\ \ 0.734)}$$ ต้องระบุของ X-1118, X-1820, X-1820ENR

7.1.3 Chamfer of Outer diameter

$$c1\_1 = \frac{(C\ upper\ limit + C\ lower\ limit)}{2}$$

$a1\_1$ is Chamfer angle of outer diameter refer customer specification

7.1.4 Chamfer of Inner diameter

$$c2\_1 = C\ maximum - (0.005 \times 25.4)$$

$a2\_1$ is Chamfer angle of inner diameter refer customer specification

7.1.5 Length of sleeve (L)

$$l1\_1 = \frac{(L + upper) + (L + lower)}{2}$$

The Sleeve with plating process should be have 2 set of tolerance for □1_1, One for the turning process and one for the after plating process. The plating process will add the plate thickness and tolerance to total length. The tolerance can be calculated by combining tolerance for turning or/and grinding and tolerance for plating.

7.1.6 Setback

$$t1\_1 = Setback\ maximum$$

7.1.7 Liner dimension

Refer PD-DV-EN-121007 and customer specification

7.2 Plating dimension

Every dimension may either have a reduced size equivalent to the thickness of the plate or may maintain the same size but with an allowance for tolerance in plate work during the turning process. It depends on the suitability of the size and tolerance specified by the customer. For instance, if the customer specifies a tolerance size of ± 0.254, the nominal value can be used in the turning process. Then, the tolerance can be reduced to ± 0.100 when it comes to the plate process. In this case, the plate thickness should be increased, and the tolerance of turning and plate should be combined.

$t2\_1 = 0.01$ or refer mean thickness plating from RE23011

7.3 Grinding dimension

7.3.1 Outer diameter (ØD)

The dimension for passivate or finish code for Aluminum sleeve material shall be referred from final Dimension. The dimension for the plating process is size allowance from RE23011 Standard for plating design tolerance.

$$d1\_2 = d1\_1 - (2 \times t2\_1)$$

7.3.2 GD&T

Provide references to GD&T for the final dimension specifically related to the dimensions specified in 8.3.1 Grinding dimension.

7.3.3 Other diameter

Other dimensions if not specified should be referred from the final dimension.

7.4 Trim and teflon bonding dimension

The dimension on trim and teflon bonding shall be referred from the final dimension. Trim process should inspect the ID chamfer, and teflon bonding should inspect both the setback length between the end of liner length and the end of sleeve length.

7.5 Turning dimension

7.5.1 Outer diameter (ØD)

$$d1\_3 = Maximum\ d1\_1 + d0$$

$$\text{Ø}B \ \leq\ \text{Ø}\,45.00 :\ d0 = 0.1$$

$$\text{Ø}B > \text{Ø}\,45.00 :\ d0 = 0.2$$

$$or\ dimension\ allowance\ for\ grinding$$

7.5.2 Chamfer dimension

$$c1\_2 = c1\_1 + \left(\frac{d1\_2 - d1\_1}{2 \times tan\ 15°}\right)$$

8. ACTION FOR NON-CONFORMING DRAWINGS

If a nonconformity of the drawing is found, after confirming the cause and examining counter measures, it is follow PD-DV-QC-230000 Control of Non-conformance

Appendix 1 Process design record sheet

Process design record

PDR Number :	
Customer :	
Procurement spec :	
Product spec :	

Record นี้จะได้ใช้ตอนไหน?

Product :	

No.	Process code	Process name	Specification Requirement Number	Public specification number	Public specification name	Revision

Notice! : This data is Reference Only

PD-DV-EN-xxxxxx

Revision control sheet

Revision	Date	Detail	Approve